

NGÂN HÀNG CÂU HỎI ÔN THI CUỐI KỲ

327 câu hỏi trắc nghiệm song ngữ Anh – Việt
8 tuần học · 22 quiz · Có đáp án ở cuối tài liệu

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Quiz01 · 12 câu

Câu 1. Which sequence correctly describes the flow of changes from your local edits to the remote server in Git?

- A. Working Directory → Local Commits → Staging Area → Remote
- B. Staging Area → Working Directory → Remote → Local Commits
- C. Working Directory → Staging Area → Local Commits → Remote
- D. Local Commits → Remote → Staging Area → Working Directory

Câu 2. Starting a New Local Project: Grace wants to start a new project in a folder on her computer named project-alpha. The project does not exist on a remote server yet. What is the very first Git command she should run inside the folder?

- A. `git clone project-alpha`
- B. `git init`
- C. `git remote add origin ...`
- D. `git commit`

Câu 3. Ignoring Generated Files: Henry is working on a project that generates log files (*.log) and a build output directory (/dist). He wants Git to completely ignore these files. What should he do?

- A. Run `git clean -fd` before every commit.
- B. Add the lines *.log and /dist to a .gitignore file.
- C. Manually delete the files before running `git add`.
- D. Use `git update-index --assume-unchanged` on the files.

Câu 4. What is the primary function of a .gitignore file?

- A. To list files that Git should permanently delete from the project history.
- B. To specify files and directories that Git should intentionally not track, such as build artifacts or local environment files.
- C. To mark certain project files as required so that Git always includes them in every commit, even when they have not been modified.
- D. To set up and manage user-wide preferences, credentials, and default behaviors for Git across all repositories on your machine.

Câu 5. The Accidental add: Ken is preparing a commit and uses `git add .` to stage everything. He then realizes he accidentally staged a sensitive file, secrets.env. How can he remove this file from the staging area while keeping it in his working directory?

- A. `git rm --cached secrets.env`
- B. `git reset secrets.env` or `git restore --staged secrets.env`
- C. `git checkout -- secrets.env`
- D. Delete the file and then restore it from the last commit.

Câu 6. What is the most significant advantage of using branches in your development workflow?

- A. They create separate, encrypted backup copies of your entire codebase that can be restored if data is lost.
- B. They reduce the amount of disk space a repository uses by storing only compressed deltas instead of full file copies on every branch.
- C. They allow you to work on new features or fixes in isolation without disrupting the stability of the main codebase.
- D. They automatically resolve merge conflicts between different developers.

Câu 7. A Forgotten File: Alice just made a commit with the message "feat: Add user login form". She immediately realizes she forgot to git add the new CSS file that styles the form. She has not pushed her commit yet. What is the most efficient way for Alice to add the missing CSS file to her last commit?

- A. Create a new commit with the message "fix: Add missing Cascading Style Sheets (CSS)".
- B. Use `git reset --hard HEAD~1` (HEAD is the current commit pointer) and then re-commit both files.
- C. Use `git add style.css` and then `git commit --amend`.
- D. Use `git revert HEAD` to undo the commit and start over.

Câu 8. What is the key difference between the `git fetch` and `git pull` commands?

- A. `git fetch` uploads your changes, while `git pull` downloads them.
- B. `git pull` is an older, deprecated command, whereas `git fetch` is the modern replacement.
- C. `git fetch` only works for the main branch, while `git pull` works for all branches.
- D. `git fetch` downloads remote updates but does not merge them, while `git pull` downloads and merges them into your current branch.

Câu 9. Which of the following best describes the "Feature branch + Pull Request" collaboration workflow?

- A. All team members push and commit their finished work directly onto the shared main branch without using separate feature branches.
- B. Work is done on a separate branch, pushed to the remote, and then a request is opened for code review before merging.
- C. A single, long-running branch is used for development, staging, and production environments.
- D. Developers frequently merge small, unreviewed changes to the main branch throughout the day.

Câu 10. Resolving a Merge Conflict: Noah is merging main into his branch, and Git reports a `MERGE_CONFLICT`. He opens the conflicted file, manually edits it to fix the conflict markers (`<<<<<`, `>>>>>`), and saves it. What are the next two commands he must run to complete the merge?

- A. `git commit` and then `git merge --continue`
- B. `git add <conflicted_file>` and then `git merge --continue`
- C. `git rebase --continue`
- D. `git status` and then `git commit`

Câu 11. What is the recommended strategy for avoiding merge conflicts on a long-running feature branch?

- A. Wait until every part of the feature is done and tested before pushing your branch to the remote repository for the first time.
- B. Frequently merge or rebase the main branch into your feature branch to stay up-to-date.
- C. Do not pull or merge any updates from the main branch into your feature branch until the feature is ready for final integration.
- D. Delete and recreate the branch from main whenever a major change happens on main.

Câu 12. Staying Synced with an Upstream Fork: Jake has forked an open-source project and cloned his fork. To contribute effectively, he needs to keep his fork updated with the original project. How should he configure his local repo to do this?

- A. Delete your local copy and run git clone on the original repository again each day to get the latest changes.
- B. Download or copy updated source files from the original project's website and paste them into your local fork by hand.
- C. Add the original project's Uniform Resource Locator (URL) as a remote named upstream.
- D. Submit his pull requests by pushing commits straight to the main branch of the original upstream repository without going through his fork.

Quiz02 · 12 câu

Câu 13. In which scenario is git revert the most appropriate command to use?

- A. When you want to undo a commit that has already been pushed to a shared branch without rewriting history.
- B. When you want to revise only the commit message of your most recent local commit that has not yet been pushed to a shared remote branch.
- C. When you need to combine the last three commits on your local branch into a single commit.
- D. When you want to permanently discard every uncommitted modification in both your working directory and your staging area.

Câu 14. What is the primary function of the git reflog command?

- A. To list every remote repository alias configured for the project along with its associated fetch and push URLs.
- B. To provide a log of where your HEAD (current commit pointer) and branch pointers have been, acting as a safety net to recover "lost" work.
- C. To show the complete commit history of the current branch in a paginated log format that can include patch diffs for each commit.
- D. To prune expired reflog entries and unreachable loose objects from the local Git repository over time so that overall disk usage on the machine is reduced.

Câu 15. Bob is in the middle of developing a complex new feature. Suddenly, a critical bug is reported in production. Bob needs to fix the bug immediately, but his current work is unfinished and not ready to be committed. What should Bob do to safely set aside his current work so he can switch branches?

- A. Run git commit -m "Work In Progress (WIP)" to save the unfinished work.
- B. Run git stash to temporarily shelve his changes.
- C. Run git reset --hard to discard his current work.
- D. Run git clean -fd to remove all untracked files.

Câu 16. When is the git stash command most useful?

- A. When you want to permanently save a version of your code as an official release.
- B. When you need to switch to another branch but have unfinished work you aren't ready to commit.
- C. When you need to remove every untracked file and directory from your working tree while leaving all tracked project files exactly as they are.
- D. When you have finished your work and are ready to push it to the remote repository.

Câu 17. Frank has finished a feature on his branch. The history contains 10 small, messy commits like "fix typo," "wip," and "try another thing." Before opening a Pull Request (PR), he wants to combine them into two meaningful commits to make the history easier to review. Which command should he use?

- A. `git merge --squash main`
- B. `git rebase -i HEAD~10`
- C. `git commit --amend` for each of the 10 commits.
- D. `git filter-branch`

Câu 18. Isabella has modified three files but has not staged them yet. She wants to see a line-by-line view of all the changes she has made in her working directory compared to the last commit. Which command will show her this?

- A. `git status`
- B. `git log -p`
- C. `git diff`
- D. `git diff --staged`

Câu 19. What information does the git blame command provide for a specific file?

- A. It shows which developer is responsible for the most bugs in that file.
- B. It provides a line-by-line breakdown of who last changed each line and in which commit.
- C. It displays a comprehensive summary of every merge conflict that has been recorded throughout the entire version history of that specific file.
- D. It generates a detailed report that describes how the file's size and modification patterns have evolved across commits over time.

Câu 20. A feature that was working in last month's release (tagged v1.5.0) is now broken on the main branch. David needs to find the exact commit that introduced this regression among the dozens of commits made since the release. What is the most efficient Git tool for this task?

- A. `git blame <file>`
- B. `git log -p`
- C. `git bisect`
- D. `git show <commit>`

Câu 21. Liam's team is ready to deploy version 2.0.0. The final commit is on main. He needs to create a permanent, formal marker for this specific commit that includes a message about the release. What should he use?

- A. `git branch v2.0.0`
- B. `git tag -a v2.0.0 -m "Version 2.0.0 release"`
- C. `git commit --allow-empty -m "Release v2.0.0"`
- D. `git note add -m "v2.0.0"`

Câu 22. Maria is trying to add a 500MB machine learning model file (model.h5) to her project. Her team uses Git Large File Storage (LFS). What is the correct procedure for her to add this file?

- A. Add model.h5 to .gitignore to keep the repo small.
- B. Run `git lfs track "*.h5"`, then `git add .gitattributes model.h5`.
- C. Commit the file normally; Git will handle it automatically.
- D. Zip the file and then commit the .zip archive.

Câu 23. How does Git Large File Storage (LFS) help manage large binary files?

- A. It prevents any individual file whose uncompressed size exceeds one hundred megabytes from ever being added to the repository's object database.
- B. It stores the content of large files on an external server, placing only lightweight pointers in the Git history.
- C. It automatically applies lossless compression to binary assets so they occupy less space inside the repository without requiring manual intervention.
- D. It converts binary file contents into a human-readable text representation so that Git's standard textual diff tools can compare versions line by line.

Câu 24. Olivia wants to see a compact, graphical view of the commit history from all branches, not just her current one. What is the best command to get this rich, condensed overview?

- A. `git log -p --all`
- B. `git shortlog --summary`
- C. `git log --oneline --graph --decorate --all`
- D. `git log --simplify-by-decoration`

Quiz01 · 18 câu

Câu 25. What is the primary goal of software testing?

- A. To execute a program with the intent of finding errors.
- B. To prove that the software has no bugs.
- C. To write as many test cases as possible.
- D. To speed up the development process.

Câu 26. Consider the following Python function:

```
def is_even(n):  
    return n % 2 == 0
```

Which of the following test cases will FAIL?

- A. `assert is_even(4) == True`
- B. `assert is_even(0) == True`
- C. `assert is_even(-2) == True`
- D. `assert is_even(3) == True.`

Câu 27. Which of the following best describes an "understanding error"?

- A. A developer makes a mistake with the upper bound of a loop, causing an "off-by-1" error.
- B. A developer misunderstands the requirements and doesn't handle an unexpected data format.
- C. A developer forgets to close a database connection, causing a performance issue.
- D. A developer uses an inefficient algorithm that slows down the program.

Câu 28. Given the following function:

```
def divide(a, b):  
    return a / b
```

Which test cases will FAIL?

- A. `assert divide(6, 2) == 3`
- B. `assert divide(5, 2) == 2.5`
- C. `assert divide(1, 0) == 0.`
- D. `assert divide(-4, 2) == -2`

Câu 29. Which type of testing is primarily concerned with whether the software works quickly and can handle the expected user load?

- A. Functional Testing
- B. User Testing
- C. Performance and Load Testing.
- D. Security Testing

Câu 30. Consider the function:

```
def get_last_item(items):  
    return items[-1]
```

Which test will FAIL?

- A. `assert get_last_item([1, 2, 3]) == 3`
- B. `assert get_last_item(['a']) == 'a'`
- C. `assert get_last_item([]) == None.`
- D. `assert get_last_item([True, False]) == False`

Câu 31. A company asks a group of its own developers and a few select users to test an early version of a product to get feedback on planned features. What is this process called?

- A. Beta Testing
- B. Release Testing
- C. Alpha Testing.
- D. Unit Testing

Câu 32. Given the following function:

```
def add(a, b):  
    return a + b
```

Fill in the missing expected value so the test verifies addition of negative numbers:

```
assert add(-2, -3) == ____
```

- A. -5

Câu 33. What is the fundamental limitation of software testing?

- A. It is too expensive and time-consuming for most projects.
- B. It can only be performed by specialized quality assurance engineers.
- C. It requires complex tools that are difficult to learn.
- D. It can demonstrate the presence of faults, but cannot prove their absence.

Câu 34. Consider the function:

```
def max_value(a, b):  
    if a > b:  
        return a  
    return b
```

Which test will FAIL?

- A. `assert max_value(3, 5) == 5`
- B. `assert max_value(10, 2) == 10`
- C. `assert max_value(4, 4) == 4`
- D. `assert max_value(-1, -2) == -2.`

Câu 35. A test is considered to "pass" under which condition?

- A. The program runs without crashing.
- B. The program's behavior matches the expected behavior.
- C. The program executes all lines of code at least once.
- D. The program is approved by the product manager.

Câu 36. Given:

```
def get_item(items, index):  
    return items[index]
```

Which test will PASS?

- A. `assert get_item([1, 2, 3], 3) == None`
- B. `assert get_item([], 0) == None`
- C. `assert get_item(['a', 'b'], 1) == 'b'.`
- D. `assert get_item([10], -2) == 10`

Câu 37. A developer creating a program to process a sequence of items makes a mistake and fails to process the very last element. This is a classic example of what?

- A. An understanding error
- B. A requirements error
- C. An "off-by-one" programming error.
- D. A system integration error

Câu 38. Which type of testing focuses on validating the functionality of the overall system and discovering as many implementation bugs as possible?

- A. User Testing
- B. Functional Testing.
- C. Performance Testing
- D. Security Testing

Câu 39. Given:

```
def is_positive(n):  
    return n > 0
```

Which assertion best tests the boundary condition?

- A. `assert is_positive(10) == True`
- B. `assert is_positive(1) == True`
- C. `assert is_positive(0) == False.`
- D. `assert is_positive(100) == True`

Câu 40. The goal of which testing type is to ensure the software maintains its integrity and can protect user information from theft and damage?

- A. User Testing
- B. Release Testing
- C. Performance Testing
- D. Security Testing.

Câu 41. Complete the test to ensure the function returns True for empty input: [\[Nhiều đáp án\]](#)

```
def is_empty(items):  
    return len(items) == 0  
  
assert is_empty([]) == ____
```

- A. True
- B. TRUE

Câu 42. When a software company distributes an early version of its product to a large, external group of users for comments on its usability in their own environments, what is this called?

- A. Alpha Testing
- B. Beta Testing.
- C. System Testing
- D. Functional Testing

Quiz02 · 15 câu

Câu 43. What is the primary goal of functional testing?

- A. To ensure the code is easy to read and maintain.
- B. To check if users find the product useful and easy to use.
- C. To test the system's response time under a heavy user load.
- D. To find bugs and provide evidence that the system is fit for its purpose.

Câu 44. Which sequence correctly orders the stages of functional testing, from the smallest component to the complete system?

- A. System testing → Feature testing → Unit testing → Release testing
- B. Unit testing → System testing → Feature testing → Release testing
- C. Unit testing → Feature testing → System testing → Release testing
- D. Feature testing → Unit testing → Release testing → System testing

Câu 45. What is the main objective of unit testing?

- A. To test that a complete user feature works as intended.
- B. To decide if the software is good enough for customers.
- C. To test program units, like functions or methods, in isolation.
- D. To test the interactions between multiple system features.

Câu 46. In the context of unit testing, which of the following best defines a "code unit"?

- A. The entire software product.
- B. A single line of code written by a developer.
- C. A specific feature requested by a user.
- D. Anything with a clearly defined responsibility, like a function or class method.

Câu 47. What is the fundamental principle behind using equivalence partitions to design tests?

- A. To divide the code into equal parts for different testers to work on.
- B. To group inputs that are treated similarly by the code, so that one test can represent the entire group.
- C. To focus exclusively on inputs that are known to have caused errors in the past.
- D. To test every single possible input to guarantee the program is completely free of faults.

Câu 48. When designing unit tests, why is it important to also identify "incorrectness partitions"?

- A. To test that the program correctly detects and handles invalid inputs as expected.
- B. To make the overall test suite smaller and faster to run.
- C. To prove that the program will never fail under any circumstances.
- D. To avoid testing edge cases that are unlikely to occur in production.

Câu 49. Feature testing is primarily concerned with which aspect of software development?

- A. Verifying that individual functions work correctly in isolation.
- B. Ensuring that integrated code units work together to provide useful functionality.
- C. Checking the system's performance and stability under heavy load.
- D. Finding security vulnerabilities that an attacker might be able to exploit.

Câu 50. Feature testing involves two types of tests: "interaction tests" and "usefulness tests." What is the main goal of usefulness tests?

- A. To check for bugs in how different code units are integrated.
- B. To verify that the feature implements something that users actually want or need.
- C. To ensure the feature's code is optimized for maximum speed.
- D. To confirm that all individual code units passed their own tests.

Câu 51. How are user stories often utilized in the feature testing process?

- A. They are converted directly into the final user documentation.
- B. They are used only for high-level system testing, not feature testing.
- C. They serve as a basis for organizing tests to ensure the feature aligns with real-world user scenarios.
- D. They are used by marketing teams to create advertisements for the product.

Câu 52. A primary goal of system testing, which is not typically covered in unit or feature testing, is to:

- A. test a single function to verify its return value.
- B. discover unexpected and unwanted interactions between different system features.
- C. check the code for proper syntax and adherence to style guides.
- D. verify the logic of an individual class method in isolation.

Câu 53. Beyond checking for feature interactions, what is another crucial objective of system testing?

- A. Ensuring every single line of code has been executed at least once.
- B. Confirming that the development team has met its deadline.
- C. Verifying that the system operates correctly in the different environments where it will be used.
- D. Making sure each developer has written enough unit tests for their code.

Câu 54. According to the provided text, what is an effective way to systematically conduct system testing?

- A. By having the original programmer test their own code without guidance.
- B. By starting with a set of scenarios describing possible user tasks and working through them.
- C. By focusing only on the newest features that have been recently added.
- D. By randomly clicking buttons and entering data to try to crash the system.

Câu 55. What is a key distinction between release testing and system testing?

- A. Release testing is automated, while system testing is always manual.
- B. Release testing focuses on finding every possible bug, while system testing focuses on performance.
- C. System testing is done in a test environment, while release testing is done in the real operational environment.
- D. System testing is performed by end-users, while release testing is performed by developers.

Câu 56. The main purpose of release testing is to:

- A. find and fix every remaining bug in the software.
- B. provide developers with feedback on code quality.
- C. decide if the system is good enough to be released to customers.
- D. test each new feature as soon as it is developed.

Câu 57. At which stage of functional testing would a developer specifically check that several code units, possibly created by different programmers, are correctly integrated to deliver a single piece of user functionality?

- A. Unit testing
- B. Feature testing
- C. System testing
- D. Release testing

Quiz03 · 15 câu

Câu 58. What is the core idea behind test automation?

- A. Tests are written as manual scripts for a human to follow.
- B. Only the user interface (UI) parts of an application are tested.
- C. Tests are written as executable code that can run without manual intervention.
- D. The goal is to prove that the software has no bugs at all.

Câu 59. The "Arrange-Action-Assert" pattern is a common structure for automated tests. What happens during the "Assert" step?

- A. Test parameters and the test environment are set up.
- B. The code unit being tested is called with specific inputs.
- C. The test checks if the actual result from the code matches the expected outcome.
- D. The results of the test are reported to the development team.

Câu 60. You are given the following test code fragments written in Python.

```
Block A
    result = add(2, 3)
```

```
Block B
    a = 2
    b = 3
```

```
Block C
    assert result == 5
```

Which option correctly orders the blocks according to the Arrange-Action-Assert (AAA) pattern?

- A. Block A → Block B → Block C
- B. Block B → Block C → Block A
- C. Block B → Block A → Block C
- D. Block C → Block A → Block B

Câu 61. Consider the following Python test fragments.

```
Block A
numbers = [1, 2, 3]

Block B
numbers.append(4)

Block C
assert len(numbers) == 4
```

Which option correctly orders the blocks according to the Arrange-Action-Assert (AAA) pattern?

- A. Block A → Block B → Block C
- B. Block B → Block A → Block C
- C. Block A → Block C → Block B
- D. Block C → Block A → Block B

Câu 62. You are given parts of a Python unit test.

```
Block A
text = "hello"

Block B
assert result == "HELLO"

Block C
result = text.upper()
```

Which option correctly orders the blocks according to the Arrange-Action-Assert (AAA) pattern?

- A. Block C → Block A → Block B
- B. Block B → Block C → Block A
- C. Block A → Block B → Block C
- D. Block A → Block C → Block B

Câu 63. What is the definition of regression testing?

- A. The process of writing new tests for a newly developed feature.
- B. The process of re-running previous tests after a change to ensure nothing has broken.
- C. The process of manually testing the software just before a major release.
- D. The process of removing old, outdated tests from the test suite.

Câu 64. What is the most significant advantage of having an automated test suite for regression testing?

- A. It eliminates the need for any manual testing whatsoever.
- B. It guarantees that no new bugs will ever be introduced into the code.
- C. It allows thousands of tests to be run in seconds, giving teams the confidence to make changes safely and quickly.
- D. It automatically generates new test cases based on user behavior.

Câu 65. What is the main principle of the Test Pyramid model proposed by Mike Cohn?

- A. To guide the structure of an automated test suite, emphasizing a large number of fast unit tests at the base.
- B. To ensure that every developer writes the same number of tests.
- C. To mandate that 100% of testing must be automated.
- D. To focus testing efforts primarily on the user interface because it's what the customer sees.

Câu 66. According to the Test Pyramid, why should you have significantly more unit tests than system/UI tests?

- A. Unit tests are more interesting for developers to write.
- B. System tests are unable to find important bugs.
- C. Unit tests are much faster, more stable, and pinpoint problems more precisely than higher-level tests.
- D. UI tests can only be performed manually and cannot be automated.

Câu 67. Which statement best describes Test-Driven Development (TDD)?

- A. A process where you write production code first and then write tests for it later.
- B. A process where you write an automated test before you write the production code needed to make it pass.
- C. A methodology where a separate testing team is solely responsible for all testing activities.
- D. A technique focused only on testing the security aspects of an application.

Câu 68. What is the correct sequence of the TDD cycle?

- A. Green → Red → Refactor
- B. Red → Refactor → Green
- C. Refactor → Red → Green
- D. Red → Green → Refactor

Câu 69. In the TDD cycle, what is the primary purpose of the "Red" phase?

- A. To write the simplest possible production code.
- B. To clean up and improve the code that was just written.
- C. To write a small, specific test that fails because the required functionality does not yet exist.
- D. To run all existing tests to make sure they are still passing.

Câu 70. After a test passes in the "Green" phase of TDD, what is the goal of the "Refactor" phase?

- A. To write another failing test for the next piece of functionality.
- B. To deploy the code immediately to production.
- C. To clean up and improve the code's design without changing its behavior, confident that the tests will catch any regressions.
- D. To ask another developer to review the code.

Câu 71. Which of the following is a key benefit of practicing TDD?

- A. It completely eliminates the need for code reviews.
- B. It guarantees that the software will be 100% bug-free.
- C. The resulting tests act as a form of executable documentation that is always up-to-date with the code.
- D. It is the fastest way to write code, especially for projects with unclear requirements.

Câu 72. According to the course materials, what is a potential challenge or downside of TDD?

- A. It results in having too few tests to cover the codebase.
- B. It can discourage developers from making radical refactoring changes if it means rewriting many tests.
- C. It simplifies debugging to the point where developers no longer learn valuable skills.
- D. It only works for user interface testing and cannot be used for backend logic.

Quiz04 · 10 câu

Câu 73. What is the primary definition of a code review?

- A. An automated process that scans code for performance issues.
- B. A formal meeting where project managers approve code for deployment.
- C. Systematic, peer-evaluation process where developers check each other's code for correctness and quality.
- D. The act of a single developer re-reading their own code before submitting it.

Câu 74. Why are code reviews a critical complement to automated testing?

- A. They are much faster than running a full suite of automated tests.
- B. They can find bugs caused by a misunderstanding of requirements, which automated tests would likely miss.
- C. They focus only on the visual aspects of the user interface.
- D. They completely eliminate the need for any further testing.

Câu 75. Besides finding bugs, what is another significant benefit of conducting regular code reviews?

- A. It helps determine which developer writes the fewest bugs.
- B. It serves as the primary way to calculate project costs.
- C. It is an excellent method for sharing knowledge about the codebase across the team.
- D. It is a substitute for writing technical documentation.

Câu 76. Which of the following issues is most effectively identified through a code review rather than by running functional tests?

- A. A calculation that produces an incorrect numerical result.
- B. A part of the program that crashes when given a null input.
- C. Code that is poorly organized and difficult to understand, even if it works correctly.
- D. A feature that doesn't meet its required response time under load.

Câu 77. According to the course materials, what should a reviewer examine in addition to the program code being reviewed?

- A. The developer's previous performance reports.
- B. The project's marketing and sales strategy.
- C. The automated tests written for the code to ensure they are complete and correct.
- D. The time it took the programmer to write the code.

Câu 78. What kind of culture should be fostered for code reviews to be successful?

- A. A highly competitive one, where reviewers try to find the most flaws to prove their expertise.
- B. A culture where reviews are seen as a way to formally evaluate and rank programmers.
- C. A positive and collaborative one, where the goal is to improve the code, not to criticize the programmer.
- D. A culture where programmers are expected to reject all feedback to show confidence in their work.

Câu 79. Which of the following is listed as a key area to look for in a code review checklist?

- A. The number of lines of code written.
- B. The time spent on the task.
- C. The correctness of the code and whether it handles edge cases.
- D. The programming language used.

Câu 80. What is the recommended best practice for the scope of a single code review session?

- A. To cover as much code as possible in a single day.
- B. To focus on a manageable amount of code (e.g., 200-400 lines) in a session lasting about an hour.
- C. To only review code written by senior developers.
- D. To continue the review until every participant agrees the code is perfect.

Câu 81. Modern code review tools often integrate with code repositories like GitHub. What is the primary benefit of this integration?

- A. It allows the tool to automatically fix all identified bugs.
- B. It can automatically set up a code review whenever a programmer submits new code.
- C. It prevents programmers from submitting code that contains any bugs.
- D. It replaces the need for any human involvement in the review process.

Câu 82. Why is a checklist a useful tool for a code reviewer?

- A. It helps the reviewer remember to check for common types of errors and ensures consistency.
- B. It serves as a legally binding contract between the programmer and the reviewer.
- C. It automatically scores the quality of the code on a scale of 1 to 10.
- D. It provides the exact code changes that need to be made.

Quiz05 · 7 câu

Câu 83. Which of the following is NOT typically part of a CI pipeline?

- A. Unit testing
- B. Code checkout
- C. Docker image build
- D. Marketing plan review

Câu 84. What is the purpose of caching dependencies in CI?

- A. To make builds slower for accuracy
- B. To skip testing
- C. To speed up repeated builds by reusing unchanged files
- D. To reduce code coverage

Câu 85. Why is containerization valuable in CI/CD pipelines?

- A. Containers ensure consistent environments across stages
- B. Containers are slower than VMs
- C. Containers remove the need for testing
- D. Containers replace all scripts

Câu 86. Which metric is part of the DORA metrics for DevOps performance?

- A. Number of developers on a project
- B. Deployment frequency
- C. Total code size
- D. Meeting hours per week

Câu 87. What is a common cause of pipeline failures in CI/CD?

- A. Too many reviewers
- B. Flaky or unstable tests
- C. Fast network connections
- D. Use of YAML syntax

Câu 88. What is the benefit of 'fail fast' in pipeline design?

- A. It hides test results
- B. It saves time by detecting errors early
- C. It ensures all jobs run regardless of failure
- D. It only applies to manual testing

Câu 89. In a CD pipeline, what does blue-green deployment help achieve?

- A. Simultaneous deployment to all servers
- B. Instant rollback and minimal downtime during releases
- C. Random user routing
- D. Cost reduction by halving infrastructure

Quiz01 · 10 câu

Câu 90. Besides preventing security attacks, what is a key reliability reason for input validation?

- A. To prevent "database pollution" from accidentally incorrect data.
- B. To make the user interface load faster.
- C. To reduce the program's memory usage.
- D. To ensure all inputs are stored in a single format.

Câu 91. What is the "whitelisting" approach to input validation?

- A. Defining a set of rules for what is not allowed (e.g., script tag).
- B. Defining a set of rules for what is allowed, and rejecting everything else.
- C. A list of users who are allowed to use the system.
- D. Allowing all inputs except those from a known bad IP address.

Câu 92. Why is whitelisting generally considered a more secure strategy than blacklisting?

- A. Blacklisting is much harder to program.
- B. Whitelisting runs faster.
- C. Attackers can often find new ways to bypass a blacklist, but it's much harder to bypass a strict whitelist.
- D. Whitelisting can be performed on the client-side, while blacklisting cannot.

Câu 93. A web form uses JavaScript to check if an email is valid before the user clicks "submit." This is...

- A. ...all the validation that is needed for a secure application.
- B. ...a form of server-side validation.
- C. ...good for user experience, but insufficient for security.
- D. ...a bad practice that slows down the user's browser.

Câu 94. For security and data integrity, where must validation always be performed?

- A. On the client-side (in the browser).
- B. On the server-side.
- C. In the database, using triggers.
- D. In a separate, dedicated validation microservice.

Câu 95. What is a "regular expression" (regex) commonly used for in validation?

- A. Encrypting the user's input before storing it.
- B. Defining a pattern (e.g., for an email or phone number) that an input string must match.
- C. Handling program crashes from bad input.
- D. Refactoring complex code.

Câu 96. When a user's input fails validation, why is it often a bad security practice to give a detailed error message (e.g., "Input is too long, must be 8 chars")?

- A. It gives attackers clues about the validation rules, helping them find a way to bypass it.
- B. It is too confusing for non-technical users.
- C. It is too difficult to translate detailed errors into other languages.
- D. It logs too much information to the server.

Câu 97. Why is it important to check that a numeric input (like 'age') is not an impossibly large number?

- A. It wastes database storage space.
- B. It is hard for humans to read.
- C. It can cause a numeric overflow exception and crash the program.
- D. It makes sorting the numbers very slow.

Câu 98. In addition to a range check (e.g., $0 < \text{age} < 120$), what is another type of "sensible value" check for numbers?

- A. A check to see if the number is odd or even.
- B. A check to convert the number to a string.
- C. A consistency check (e.g., ensuring a new meter reading is higher than the previous one).
- D. A check to see if the number is a prime number.

Câu 99. In many regex dialects, what does the \$ (dollar sign) symbol mean when used at the end of an expression?

- A. The match must end at the end of the string.
- B. The previous character is optional.
- C. It matches a literal dollar sign.
- D. It matches any single number.

Quiz02 · 12 câu

Câu 100. What is the primary goal of failure management?

- A. To guarantee that the program has zero bugs in every release before it reaches users.
- B. To minimize the impact of failures on the user and prevent data loss.
- C. To automatically identify, repair, and redeploy fixes for all bugs as soon as they are found.
- D. To make the program run faster so users experience fewer delays and therefore fewer failures.

Câu 101. The principle of "failing secure" means that a program failure should...

- A. ...never happen under any normal operating condition, regardless of inputs or environment.
- B. ...be hidden from the user so they are not alarmed and the system appears to continue normally.
- C. ...securely delete all user data immediately so no attacker can read it after the failure.
- D. ...never leave confidential data in an exposed or insecure state.

Câu 102. A program crashes because it tries to open a file that does not exist. This is best described as a...

- A. Program exception.
- B. Data failure caused by invalid or corrupted input rather than the program's inability to continue.
- C. Timing failure caused by the operation taking too long or completing in the wrong order.
- D. Hardware failure caused by a physical storage device or network component malfunctioning.

Câu 103. When updating a bank database, you must (1) debit account A and (2) credit account B. What mechanism ensures that both steps complete, or neither does?

- A. An if-else block that checks each account balance and manually decides which assignment should run
- B. A try-except block that catches any database error and then continues with the remaining step
- C. An ACID Transaction
- D. A regular expression used to validate the account numbers before either balance is changed

Câu 104. What is the main purpose of an exception-handling mechanism like try-except?

- A. To "catch" runtime errors and transfer control to a special block of code for graceful cleanup.
- B. To validate user input before it is processed so invalid form values never reach the main code path during execution.
- C. To prevent errors from ever happening in the first place by blocking every possible runtime fault before execution begins.
- D. To automatically repair the underlying bug, rebuild the program, and retry the failed operation without developer involvement.

Câu 105. If a function catches an exception, performs a local cleanup (like closing a file), and then re-raises the exception, why would it do this?

- A. To log the exception to the console twice.
- B. To crash the program intentionally.
- C. To allow a higher-level (calling) function to also handle the exception.
- D. To prove to the compiler that the exception was caught and therefore no higher-level error handling is needed.

Câu 106. What are two common mechanisms to help a user recover work after a program crash?

- A. Auto-save and Activity logging.
- B. Error messages and Pop-up alerts that notify the user about the crash and describe what may have happened.
- C. Unit testing and Integration testing performed during development to reduce defects before the application is released.
- D. Whitelisting and Blacklisting used to control which inputs, addresses, or actions are allowed by the system.

Câu 107. Why do external web APIs typically return numeric error codes (like HTTP 404 or 500) instead of language-specific exceptions?

- A. The API and the calling program may be written in different, incompatible programming languages.
- B. Numbers are faster to send over the internet because they use less bandwidth than descriptive messages or exception objects.
- C. Numeric codes are more secure because attackers cannot intercept, read, or understand short numerical responses during communication.
- D. It is a tradition that has no real technical purpose.

Câu 108. What is the simplest way to prevent your application from hanging forever if an external API call never responds?

- A. Use a timeout mechanism.
- B. Restart the server before making each request so any stuck connection is cleared ahead of time.
- C. Send the request again immediately.
- D. Call a different API whenever the first service is slow, even if it provides unrelated data.

Câu 109. What design pattern is used to handle a repeatedly failing external service by "tripping" and temporarily blocking all new requests to it?

- A. The Circuit Breaker Pattern
- B. The Observer Pattern, which lets objects subscribe to notifications when another object changes
- C. The Facade Pattern
- D. The Transaction Pattern, which groups operations so they commit together or roll back as a unit

Câu 110. What is the purpose of an assert statement?

- A. To handle a known program exception, like try-catch, and recover gracefully from expected runtime errors.
- B. To validate user input from a form before the program accepts the values and continues processing.
- C. To check if a condition is true (as a "sanity check") and raise an error if it's false.
- D. To print a debug message to the console whenever a developer wants to inspect a variable value.

Câu 111. When your program has an unrecoverable internal crash, what is the best practice for the message shown to the user?

- A. A detailed technical error log (e.g., "NullPointerException at line 42") with stack-trace details for debugging.
- B. A simple, non-technical message (e.g., "An unexpected error occurred").
- C. No message at all (just close the application) so the user is not interrupted by confusing information.
- D. A message blaming the user for the error and instructing them to avoid causing the crash again.

Quiz01 · 17 câu

Câu 112. What is the primary goal of defining a software architecture?

- A. To choose the specific programming language, database engine, and low-level library stack before understanding the wider structure.
- B. To document the system's high-level structure and component interactions.
- C. To write the first modules of code for the application.
- D. To finalize the user interface (UI) design, visual style, and branding before clarifying system responsibilities.

Câu 113. A software system's quality attributes (like performance, reliability, and security) are also known as:

- A. Detailed functional features requested by end users
- B. Non-functional requirements
- C. Business goals
- D. User stories

Câu 114. A non-functional requirement states, "The system must be able to handle a 10x increase in user traffic over one year." This requirement directly influences which quality attribute?

- A. Usability of the product interface during normal navigation
- B. Security against unauthorized access and data leaks
- C. Scalability
- D. Maintainability

Câu 115. Why is it critically important to get the architecture right early in the project?

- A. Because good architecture is cheap and easy to implement once the team selects modern frameworks and tools.
- B. Because poor architectural decisions are extremely difficult and expensive to change later.
- C. Because the architecture determines the final price of the product without needing further market or user analysis.
- D. Because the first design is always the best one when created before coding begins and should not be revisited later.

Câu 116. (Scenario) A company builds an e-commerce site. As it becomes popular, the site crashes during sales. This system is failing which quality attribute?

- A. Scalability
- B. Usability of the product navigation and checkout screens
- C. Security
- D. Maintainability

Câu 117. Which quality attribute refers to the ease with which developers can fix bugs or add new features?

- A. Performance of request processing under heavy load
- B. Resilience during partial infrastructure failure
- C. Maintainability
- D. Availability

Câu 118. The process of architectural design is best described as:

- A. A series of trade-offs to balance conflicting goals
- B. A fixed checklist of components to include regardless of project goals, constraints, and quality attribute priorities
- C. A task only for the most senior developer
- D. A final step completed just before release

Câu 119. What is the most common trade-off when implementing very strict and complex security measures?

- A. The system becomes less maintainable because every security rule automatically makes all future code changes unsafe.
- B. The system's performance improves because encryption and extra authentication always reduce server workload.
- C. The system becomes less usable for the end-user.
- D. The system becomes cheaper to build.

Câu 120. What is the most common trade-off when breaking a large system into many small, independent services to improve maintainability?

- A. The system's overall performance may decrease.
- B. The system becomes less secure.
- C. The system becomes harder to scale because smaller independent services cannot be deployed separately.
- D. The system's user interface becomes more complex.

Câu 121. Which of the following is an organizational (human) factor that influences architectural decisions?

- A. The type of database selected after considering storage, query, and operational constraints
- B. The speed of the server's CPU when measured without considering team communication or ownership
- C. The skill level and experience of the development team
- D. The network protocol being used by clients and servers during normal runtime communication

Câu 122. Which quality attribute refers to a system's ability to continue operating even if some of its components have failed?

- A. Resilience (or Fault Tolerance)
- B. Performance under normal load when every component remains healthy
- C. Usability
- D. Security against unauthorized access and data breaches

Câu 123. A design with a single, central, highly-protected "vault" for all its data is:

- A. Highly resilient but complex to manage because all failures can be isolated without affecting the central store.
- B. Simple to protect, but has a single point of failure.
- C. The best design for all high-security systems.
- D. Very fast but not secure.

Câu 124. How is high availability (like "99.999% uptime") typically achieved in a system's architecture?

- A. By using a faster programming language so every request completes before any server can fail
- B. By implementing redundancy for critical components
- C. By making the user interface simpler
- D. By hiring more developers to manually restart failed services whenever outages happen

Câu 125. What is the main disadvantage of building a system with high redundancy?

- A. It makes the system less secure because adding backup components always removes authentication and authorization.
- B. It makes the system slower for users.
- C. It significantly increases the system's cost and complexity.
- D. It makes the system impossible to maintain.

Câu 126. What is the primary goal of creating architectural diagrams and documents?

- A. To create a legal contract for the client that replaces discussion between engineers and stakeholders
- B. To build a shared understanding among the development team
- C. To prove the system will have no bugs before any testing, implementation, or review is performed
- D. To automatically generate all the code without developers needing to make design decisions

Câu 127. Which of the following is NOT a non-functional quality attribute?

- A. A user must be able to reset their password.
- B. The system must be secure from common attacks during normal public internet use.
- C. The system must respond to user queries in under 2 seconds even during regular peak usage.
- D. The system must be online 24/7.

Câu 128. Which of the following is a key influence on architectural design?

- A. The expected lifetime of the product
- B. The brand and logo of the product as chosen by the marketing team before implementation
- C. The marketing campaign for the product
- D. The individual names of the developers

Quiz02 · 10 câu

Câu 129. Which architectural pattern separates an application into three interconnected components: data, user interface, and business logic?

- A. Model-View-Controller (MVC)
- B. Client-Server with request/response communication roles between clients and centralized servers
- C. Layered Architecture with presentation, business, and data access tiers arranged vertically
- D. Microservices

Câu 130. In a Layered Architecture, what is the 'golden rule' of communication between layers?

- A. Any layer can call any other layer whenever it needs data or functionality
- B. A layer can freely skip intermediate layers when direct access seems more convenient
- C. A layer can only call the layer immediately adjacent to it (above or below).
- D. A layer can only call services in the layer immediately below it.

Câu 131. What is a primary disadvantage of a monolithic architecture compared to a microservices architecture?

- A. They are simpler to deploy and test in the beginning because all modules are packaged together
- B. The entire application must be redeployed even for a small change.
- C. They are generally more secure by default without needing additional design decisions
- D. They have lower network latency for internal communication because calls stay inside one process

Câu 132. Which is a cross-cutting concern?

- A. Coupon calculation for one specific checkout promotion rule
- B. Search ranking inside a particular product catalog feature
- C. Security (e.g., auth, protection)
- D. Image thumbnail size used only on a single gallery screen

Câu 133. In MVC for web UIs, which statement is accurate?

- A. Model lives only on clients and stores temporary screen state for a single page
- B. Views register for model changes and refresh when the model updates
- C. Controllers only render static HTML pages without processing user actions or input events
- D. Servers never update models

Câu 134. Which statement about HTTP in client-server apps is correct?

- A. Servers initiate requests to clients by default before any browser action occurs
- B. Clients send requests; servers respond
- C. HTTP is binary only and cannot carry readable request or response headers
- D. HTTP requires XML payloads for every request and every response message

Câu 135. Why is JSON commonly preferred over XML here?

- A. Browsers can't parse XML documents correctly in modern web applications because only JSON is supported by JavaScript
- B. JSON is usually more compact and faster to parse, and maps to objects easily
- C. JSON guarantees ACID transaction behavior for database operations across distributed microservice databases
- D. XML can't encode numbers or structured text values in a reliable way for browser-to-server communication

Câu 136. In a multi-tier web architecture, which pairing is correct?

- A. Web server - user authentication and authorization logic only
- B. Application server - application-specific operations
- C. Database server - HTML page generation for browser rendering
- D. Web server - ACID transaction processing for persistent data updates

Câu 137. What's the main role of a load balancer in client-server systems?

- A. Encrypt database tables before storing sensitive information permanently for every application server request
- B. Distribute requests across servers to share load and improve availability
- C. Replace authentication and authorization services for logged-in users across all backend applications
- D. Eliminate all network latency between clients, application servers, and databases in every network path

Câu 138. In the “multiple database per component” design, what extra piece keeps shared data correct?

- A. A CDN that caches static images, scripts, and stylesheets near users
- B. A reconciliation component/process
- C. A sticky-session cookie that keeps one user connected to the same server
- D. A single global mutex that locks every distributed database during updates

Quiz01 · 11 câu

Câu 139. Why is it beneficial for services in a cloud-based system to be stateless?

- A. Stateless services are easier to program and require less code.
- B. They can be easily replicated, run in parallel, and moved between virtual servers to handle scaling.
- C. They are the only type of service that can communicate using XML-based protocols.
- D. They maintain a persistent, in-memory copy of the user's session, which improves performance.

Câu 140. What was a significant problem with the "Web Services" standards (e.g., SOAP, WSDL) from the early 2000s?

- A. They were too simple and did not support complex business logic.
- B. They could only be written in one programming language.
- C. They involved large, complex XML messages that were slow to analyze and had high overhead.
- D. They did not have a standard for interface description.

Câu 141. What key insight, pioneered by companies like Amazon, helped define the modern microservice?

- A. Services should be completely independent, with their own database and their own user interface.
- B. All services in a system should share a single, highly optimized database to reduce complexity.
- C. Services should focus only on technical tasks, while a central monolith handles all business logic.
- D. A service should be related to multiple business functions to be efficient.

Câu 142. Which of the following describes the ideal design goal for microservices?

- A. High coupling and high cohesion
- B. Low coupling and low cohesion
- C. High coupling and low cohesion
- D. Low coupling and high cohesion

Câu 143. Why is low coupling an important characteristic for microservices?

- A. It ensures that all services can be written in the same programming language.
- B. It allows you to update a service (while maintaining its interface) without having to change other services in the system.
- C. It minimizes the number of functions inside a single service, making the code simpler.
- D. It forces all services to share a single, common database, which simplifies data management.

Câu 144. When determining the appropriate "size" for a microservice, what guideline does the author suggest?

- A. The service's code should not exceed a strict limit, such as 500 lines of code.
- B. The service should be small enough to be developed, tested, and deployed by a team in two weeks or less.
- C. The service must encapsulate at least five related business functions to be considered a "microservice."
- D. The service should be large enough to handle a complete, end-to-end business workflow (e.g., user registration).

Câu 145. What is the "common closure principle" guideline for decomposing a system?

- A. All services that are part of the same business capability should be grouped together.
- B. Elements of a system that are likely to be changed at the same time should be located within the same service.
- C. All services should be "closed" for modification but "open" for extension.
- D. Services should be decomposed based on the data they manage, not their function.

Câu 146. What is the "orchestration" approach to service coordination?

- A. Services communicate by publishing events that other services subscribe to, with no central controller.
- B. A separate controller service (the "conductor") executes the workflow by calling component services in a specific order.
- C. All services share a central database, which orchestrates the workflow using ACID transactions.
- D. Services directly call each other in a peer-to-peer mesh, negotiating the workflow steps as they go.

Câu 147. What is a "compensating transaction"?

- A. A transaction that bundles all data updates into a single unit to guarantee consistency.
- B. A transaction that reverses a previous operation when a failure is detected in a dependent service.
- C. A log of all pending updates used to ensure "eventual consistency" between service replicas.
- D. A special transaction used by a circuit breaker to test if a failed service has recovered.

Câu 148. How does the "choreography" approach to service coordination work?

- A. A central controller dictates the exact order of operations for all services.
- B. Each service emits an event to indicate it has completed processing, and other services react to those events.
- C. The API gateway is responsible for coordinating all service workflows.
- D. Services use synchronous, direct calls to coordinate their actions in real-time.

Câu 149. Why is the "circuit breaker" pattern often preferred over simple timeouts for managing external service failures?

- A. A circuit breaker encrypts the service request, making it more secure.
- B. A circuit breaker immediately denies access to a failed service without waiting for the timeout, preventing system slowdown.
- C. A circuit breaker automatically retries the request dozens of times until the service responds.
- D. A circuit breaker automatically scales up a new instance of the failed service.

Quiz02 · 9 câu

Câu 150. What does the acronym REST stand for?

- A. Remote Execution Strategy for calling server-side procedures across a network
- B. REpresentational State Transfer
- C. Resource Entity Service Transfer for moving object records between distributed services
- D. Reliable Enterprise Service Transport

Câu 151. What is the fundamental concept of a "resource" in the REST architectural style?

- A. A function or method call that can be executed remotely through an endpoint, where the URL names an action rather than an addressable domain object.
- B. Any chunk of data (e.g., a database record, document, or catalog item) that is identified by a unique URI.
- C. A standardized message protocol, like SOAP or AMQP.
- D. A long-lived stateful connection that is maintained between the client and server so each later request depends on previous session context.

Câu 152. In the RESTful architectural style, which HTTP verb is correctly mapped to its intended operation for managing a resource?

- A. GET is used to create a new resource when the client sends parameters in the query string.
- B. POST is used to modify an existing resource's data.
- C. PUT is used to update or modify an existing resource.
- D. DELETE is used to retrieve a read-only representation of a resource.

Câu 153. How should the GET verb be used in a RESTful service?

- A. To create a new resource with a server-defined URI while also changing collection state each time the same request is sent.
- B. To read a resource, and it should never update or change the resource's value.
- C. To update an existing resource at a specific URI.
- D. To delete all sub-resources under a specific URI.

Câu 154. Which HTTP verb is used to create a new resource, with the server typically assigning the new resource's URI?

- A. GET, because retrieving the collection endpoint can also create a new item automatically.
- B. PUT, because the client asks the server to choose a fresh URI for the new item.
- C. POST
- D. CREATE

Câu 155. According to the text, what is the most commonly used representation for digital resources in modern RESTful microservices?

- A. XML, especially because modern microservices usually require verbose schema-first payloads.
- B. HTML, because browser-oriented page markup is normally used as the API payload format.
- C. Plain Text, since REST APIs generally avoid structured key-value representations.
- D. JSON

Câu 156. In an HTTP request for a RESTful service, what is the purpose of the Content-Type header?

- A. It specifies the data format (e.g., application/json) that the client is sending in the request body.
- B. It specifies the content-types (e.g., application/json) that the client can process and finds acceptable in the service response.
- C. It specifies the exact length and encoding of the response body so the client can decide which parser should be used.
- D. It provides the authentication token to prove the client is allowed to access the resource.

Câu 157. In an HTTP response, what does the status code 200 indicate?

- A. The request has failed because the requested resource was not found, and the response body normally contains the missing resource representation.
- B. The request has been successfully processed, but there is no associated content to return.
- C. The request has been successfully processed, and the response body typically contains the requested data.
- D. The request has failed because the input data was malformed (e.g., "Unprocessable entity").

Câu 158. What does the HTTP status code 204 signify?

- A. The resource was not found, but the operation is still considered partially successful because the server accepted the request.
- B. The request was successful, but there is no content to return.
- C. The service has encountered an internal failure, but it is still returning an empty success response to avoid exposing server details.
- D. The request is malformed, and the server successfully applied the invalid changes without returning any representation.

Quiz03 · 10 câu

Câu 159. What is the "DevOps" approach to service management?

- A. A separate "operations" team is responsible for deploying and managing all services created by the "development" team.
- B. The development team is responsible for deployment and service management as well as software development.
- C. It is an approach where all development and operations are outsourced to a third-party vendor.
- D. It is a set of tools used to automate only the testing, but not the deployment, of services.

Câu 160. What is the core principle of "continuous deployment"?

- A. New versions of the software are bundled and released periodically, typically three or four times per year.
- B. As soon as a change to a service has been made and validated, the modified service is automatically re-deployed into production.
- C. Developers are responsible for manually deploying their own services after getting approval from the operations team.
- D. All microservices in the system must be deployed at the same time to ensure compatibility.

Câu 161. According to the continuous deployment pipeline diagram, what event triggers the start of the automated process?

- A. A service monitoring system detects a performance failure.
- B. A user submits a new feature request to the development team.
- C. A programmer commits a code change to the version management system (e.g., Git).
- D. The team manager approves the deployment in a weekly planning meeting.

Câu 162. In the continuous deployment pipeline, what is the primary purpose of a "container" (e.g., Docker)?

- A. To act as a virtual server that includes a full operating system.
- B. To be a virtualized environment (deployment unit) that includes all the software and dependencies that a service needs to run.
- C. To run the automated unit tests before building the system.
- D. To monitor the service for failures after it has been deployed.

Câu 163. What is the role of a container management system like Kubernetes?

- A. It is a version control system used to store the source code for a service.
- B. It is a tool for writing and running automated unit tests.
- C. It automates container deployment and management, handling scheduling, load balancing, and service discovery.
- D. It is a type of API gateway used to route traffic to different service versions.

Câu 164. In the continuous deployment pipeline, which automated tests are run after the service is containerized and deployed?

- A. Unit tests
- B. Integration tests
- C. Acceptance tests
- D. All of the above

Câu 165. How can a system using an API gateway safely manage a "rollback" if a newly deployed service version fails?

- A. The API gateway automatically deletes the failed service container and rebuilds the old one from source code.
- B. The system maintains the old version, and if the monitor detects a problem, it simply switches the "current version link" back to the older service.
- C. All client applications are forced to update their URIs to point to the older service version.
- D. The system blocks all new requests, forcing them to time out until a developer manually fixes the bug in the new version.

Câu 166. When a service's API must change, how can the system support both old and new clients?

- A. By forcing all client services to update at the exact same time.
- B. By including the version number as part of the resource's URI.
- C. By deploying the new service on a completely different server domain.
- D. By using a "compensating transaction" to automatically translate new API calls to the old format.

Câu 167. What is the purpose of a numbering convention like changing a service version from 001 to 101?

- A. To indicate a minor bug fix that does not change the API.
- B. To signal to the API gateway that this service should be deployed in a container.
- C. To identify whether a new version has introduced an API change.
- D. To show that the service is now managed by a different development team.

Câu 168. According to the text, what is a primary risk of deploying new software services that testing can never completely eliminate?

- A. The cost of the cloud servers will increase.
- B. Unanticipated problems will be caused by the interactions between the new version and existing services.
- C. The service's source code will be stolen during deployment.
- D. The service will be deployed using the wrong programming language.

Quiz04 - 40 câu

Câu 169. What is the main idea of microservices architecture?

- A. Build one large application with all features inside
- B. Split an application into small independent services
- C. Store all data in one shared database
- D. Deploy only frontend applications

Câu 170. Which is a common advantage of microservices?

- A. Easier distributed debugging
- B. No need for deployment planning
- C. Independent service deployment
- D. One team must own all features, databases, and operational infrastructure for the entire company system

Câu 171. A company has one large codebase where a small change requires redeploying the whole system. What architecture is this likely?

- A. Microservices
- B. Serverless
- C. Monolithic
- D. Event-driven

Câu 172. What is a major disadvantage of microservices?

- A. Too simple to operate
- B. Increased distributed system complexity
- C. No need for monitoring
- D. Cannot use databases

Câu 173. What does loose coupling mean in microservices?

- A. Services depend heavily on each other's internal code
- B. Services communicate through clear interfaces
- C. Services share the same source files
- D. Services must use the same programming language

Câu 174. Why might different teams own different microservices?

- A. To reduce responsibility
- B. To allow independent development and delivery
- C. To force all teams to use one database
- D. To avoid testing

Câu 175. Which protocol is commonly used for public HTTP APIs?

- A. REST
- B. BIOS
- C. SMTP only
- D. FTP only

Câu 176. Which communication style is usually better when services should not wait for each other immediately?

- A. Synchronous REST call
- B. Asynchronous messaging
- C. Direct database sharing between multiple unrelated backend services
- D. Manual file copy

Câu 177. What is the role of an API Gateway?

- A. Store all business data
- B. Act as an entry point for client requests
- C. Replace all backend services
- D. Compile source code

Câu 178. A payment service calls an order service and waits for an immediate response. What type of communication is this?

- A. Asynchronous communication using event queues and delayed processing pipelines
- B. Synchronous
- C. Offline batch processing
- D. Broadcast-only communication

Câu 179. Which technology is commonly used as a message broker?

- A. RabbitMQ
- B. Microsoft Word
- C. Photoshop
- D. GitHub README

Câu 180. Why should microservices avoid depending too much on synchronous calls?

- A. It makes frontend design impossible
- B. It can create cascading failures
- C. It removes all databases
- D. It prevents using Docker

Câu 181. Why is Docker useful for microservices?

- A. It removes the need for testing
- B. It packages an application with its dependencies
- C. It replaces all databases automatically in cloud-native systems
- D. It only works for frontend apps

Câu 182. What is Kubernetes mainly used for?

- A. Writing Java code
- B. Managing and orchestrating containers
- C. Designing UI icons
- D. Creating exam questions

Câu 183. What does horizontal scaling mean?

- A. Increasing CPU and memory of a single server continuously over time
- B. Adding more service instances
- C. Deleting old logs
- D. Moving code into one file

Câu 184. What is a rolling update?

- A. Updating all services manually once per year
- B. Gradually replacing old instances with new ones
- C. Deleting production before deployment
- D. Deploying without testing

Câu 185. What is CI/CD?

- A. A database model
- B. Continuous Integration and Continuous Delivery or Deployment
- C. A network cable type
- D. A testing-only tool

Câu 186. A service has high traffic during registration week. What should the operation team likely do?

- A. Shut down the service
- B. Scale the service horizontally
- C. Merge all services into a larger monolithic application before peak registration traffic
- D. Disable monitoring

Câu 187. What is the database-per-service pattern?

- A. Every service owns its own database
- B. All services must share one database
- C. Only frontend services use databases
- D. Databases are not allowed

Câu 188. Why is a shared database risky in microservices?

- A. It makes services too independent
- B. It creates tight coupling between services
- C. It prevents SQL queries
- D. It improves all security automatically

Câu 189. What does eventual consistency mean?

- A. Data synchronization permanently fails between services
- B. Data may become consistent after some delay
- C. Data is always instantly synchronized across every service and database
- D. Data cannot be updated

Câu 190. Which pattern helps manage transactions across multiple services?

- A. Singleton
- B. Saga
- C. Factory Method
- D. MVC

Câu 191. What is a circuit breaker used for?

- A. Stop repeated calls to a failing service
- B. Encrypt database passwords
- C. Build container images
- D. Design user interfaces

Câu 192. Why are timeouts important in microservices?

- A. They make services wait forever
- B. They prevent a service from waiting too long
- C. They remove the need for APIs, monitoring systems, and distributed tracing tools
- D. They replace monitoring

Câu 193. What problem can retries cause if used carelessly?

- A. Reduced network traffic
- B. Increased load on a failing service
- C. Better UI design
- D. Automatic database migration

Câu 194. What is fault isolation?

- A. Allowing one failed service to crash the whole system
- B. Preventing one service failure from spreading
- C. Removing all error handling
- D. Keeping all code in one service

Câu 195. A recommendation service fails, but users can still place orders. What principle is shown?

- A. Total system failure caused by tightly coupled service dependencies
- B. Graceful degradation
- C. Manual deployment coordination
- D. Database sharing

Câu 196. What is load balancing?

- A. Sending all requests to one instance
- B. Distributing requests across multiple instances
- C. Deleting unused services
- D. Combining all APIs into one database

Câu 197. Why is centralized logging useful?

- A. It hides errors
- B. It collects logs from many services in one place
- C. It removes the need for testing
- D. It prevents deployment

Câu 198. What does distributed tracing help developers understand?

- A. How a request travels across services
- B. How to write HTML only
- C. How to remove APIs
- D. How to design database tables only

Câu 199. Which tool is commonly used for metrics monitoring?

- A. Prometheus
- B. PowerPoint presentation software used for creating engineering management reports
- C. Notepad
- D. Excel only

Câu 200. A user request is slow, but no single service clearly fails. What observability method is most helpful?

- A. Distributed tracing
- B. Ignoring the issue
- C. Manual screenshot
- D. Reinstalling the frontend

Câu 201. What is JWT commonly used for?

- A. User authentication and authorization tokens
- B. Container orchestration across distributed Kubernetes environments
- C. Database indexing
- D. UI animation

Câu 202. Why should services use HTTPS/TLS?

- A. To make requests slower
- B. To encrypt communication
- C. To remove authentication
- D. To replace logs

Câu 203. Where is authentication often handled in a microservices system?

- A. API Gateway or identity service
- B. CSS file
- C. Local text document
- D. User browser only

Câu 204. Why is service-to-service authentication important?

- A. Internal services should blindly trust all requests
- B. Services need to verify other services
- C. It removes all network traffic
- D. It replaces deployment pipelines

Câu 205. Why is API versioning important?

- A. To allow changes without breaking existing clients
- B. To delete old services immediately after every deployment cycle
- C. To avoid documentation
- D. To prevent testing

Câu 206. A team changes an API response format and another team's service fails. What was likely missing?

- A. UI animation
- B. Backward compatibility or contract testing
- C. More comments in CSS
- D. A shared password

Câu 207. In professional microservices development, who should usually own a service?

- A. A clear responsible team
- B. No one
- C. Only the database administrator
- D. Only the customer

Câu 208. What does DevOps culture encourage?

- A. Separation between developers and operations with no communication
- B. Collaboration between development and operations
- C. Manual deployment only
- D. Avoiding monitoring

Quiz01 · 11 câu

Câu 209. What is the main goal of Requirement Engineering (RE)?

- A. To design the software architecture
- B. To identify, analyze, and document what the system should do
- C. To write and test program code
- D. To deploy the final software

Câu 210. Which of the following best describes a “requirement”?

- A. A wish list from users
- B. A measurable statement of what the system must do or a constraint it must satisfy
- C. A design diagram of the system
- D. A testing plan for the final product

Câu 211. What is the biggest risk of skipping the RE phase?

- A. Increased speed of development
- B. Misunderstanding stakeholder needs leading to project failure
- C. Lack of programming resources
- D. Poor user interface design only

Câu 212. Which activity does NOT belong to Requirement Engineering?

- A. Eliciting stakeholder needs
- B. Defining software requirements
- C. Writing code in Python
- D. Validating requirements with users

Câu 213. Most software project failures are primarily due to:

- A. Lack of testing
- B. Poor requirements and changing needs
- C. Low coding standards
- D. Poor hardware selection

Câu 214. The phrase “System shall ...” is typically used to express:

- A. Design choices
- B. Implementation tasks
- C. Requirements statements
- D. Testing criteria

Câu 215. The primary purpose of RE is to:

- A. Gather user needs and define system behavior
- B. Implement the software solution
- C. Maintain servers
- D. Write design patterns

Câu 216. In the SDLC, RE is positioned:

- A. Before design and coding
- B. After system deployment
- C. Parallel to maintenance only
- D. During debugging

Câu 217. A “stakeholder” in RE is:

- A. Only the end-user
- B. Anyone affected by or interested in the system
- C. The developer team only
- D. Only project managers

Câu 218. What is the main difference between a requirement and a design?

- A. Requirements define what; designs define how
- B. Requirements define how; designs define what
- C. They are identical in purpose
- D. Designs are always written by clients

Câu 219. In Requirement Engineering, “scope creep” happens when:

- A. Requirements are reduced due to constraints
- B. New requirements are added without control
- C. Users are removed from the process
- D. The project is completed early

Quiz02 · 12 câu

Câu 220. Functional requirements describe:

- A. System constraints or performance expectations
- B. What the system should do
- C. The physical network structure
- D. Organizational policies only

Câu 221. Which of the following is an example of a functional requirement?

- A. System shall support 500 users simultaneously.
- B. The interface must be intuitive.
- C. System shall allow mentors to view student profiles.
- D. Response time must be under 3 seconds.

Câu 222. Which statement represents a non-functional requirement?

- A. Allow mentors to upload a profile picture.
- B. Encrypt all user data during transmission.
- C. Allow students to log in via university ID.
- D. Enable admins to reset passwords.

Câu 223. Non-functional requirements are also known as:

- A. System constraints
- B. Quality attributes
- C. Data models
- D. Prototypes

Câu 224. The “FURPS” model includes all except:

- A. Functionality
- B. Usability
- C. Reliability
- D. Productivity

Câu 225. Which example is a domain requirement for a university system?

- A. Must store mentor contact information.
- B. Must integrate with university login system.
- C. Should display mentor list within 3 seconds.
- D. Must allow password reset feature.

Câu 226. Why are domain requirements important?

- A. They define visual design rules.
- B. They prevent violation of external constraints and regulations.
- C. They help reduce code duplication.
- D. They determine testing order.

Câu 227. Which requirement is poorly written?

- A. System shall display mentor list.
- B. System shall be user-friendly.
- C. System shall validate student ID.
- D. System shall log mentor sessions.

Câu 228. Functional and non-functional requirements differ because:

- A. Functional define features; non-functional define qualities.
- B. Both describe coding logic.
- C. Non-functional are optional.
- D. Functional depend on the hardware.

Câu 229. In testing, non-functional requirements are verified by:

- A. Functional unit tests
- B. Measurement and performance evaluation
- C. System walkthroughs only
- D. Code review

Câu 230. Which type of requirement usually originates from laws or industry regulations?

- A. Functional
- B. Domain
- C. Non-functional
- D. Performance

Câu 231. Which of the following sets best represents the three types of requirements?

- A. Behavioral, System, Logical
- B. Functional, Non-functional, Domain
- C. Physical, Logical, Conceptual
- D. Performance, Usability, Design

Quiz03 · 12 câu

Câu 232. Which of the following lists the main phases of RE in order?

- A. Specification → Validation → Elicitation → Management
- B. Elicitation → Analysis → Specification → Validation → Management
- C. Analysis → Elicitation → Validation → Specification
- D. Planning → Implementation → Testing

Câu 233. The goal of the elicitation phase is to:

- A. Code the software
- B. Identify and gather stakeholder needs
- C. Write test scripts
- D. Produce user manuals

Câu 234. Which technique is commonly used during requirement elicitation?

- A. Unit testing
- B. Interviews
- C. System integration
- D. Debugging

Câu 235. What is a common challenge in elicitation?

- A. Stakeholders rarely provide conflicting requirements.
- B. Users easily express all their needs.
- C. Hidden assumptions and vague goals
- D. Developers don't participate.

Câu 236. The analysis phase focuses on:

- A. Writing final code
- B. Resolving requirement conflicts and prioritizing needs
- C. Conducting security audits
- D. Defining test metrics

Câu 237. The MoSCoW method is used for:

- A. Designing the UI
- B. Requirement prioritization
- C. Project budgeting
- D. Team scheduling

Câu 238. The output of the specification phase is typically a:

- A. Source code file
- B. Software Requirements Specification (SRS)
- C. UML activity diagram
- D. Test execution plan

Câu 239. Requirement validation ensures that requirements are:

- A. Implemented efficiently
- B. Correct, complete, consistent, and feasible
- C. Prioritized by management only
- D. Frozen after documentation

Câu 240. Which method can be used to validate requirements?

- A. Unit tests
- B. Walkthroughs and reviews
- C. Deployment reports
- D. Database normalization

Câu 241. Requirement management is necessary because:

- A. Requirements never change
- B. It handles updates, traceability, and version control
- C. It only tracks test cases
- D. It focuses on team structure

Câu 242. What tool could support requirement management?

- A. Visual Studio Code
- B. Jira or ReqView
- C. MySQL
- D. Git Bash

Câu 243. In Agile methods, Requirement Engineering is:

- A. Done once before coding
- B. Continuous through backlog refinement and iteration
- C. Skipped entirely
- D. Done after deployment

Quiz01 · 10 câu

Câu 244. The primary purpose of a software process model is to:

- A. Force all developers to work the same way.
- B. Provide structure and guidance to the software development lifecycle.
- C. Remove the need for customer involvement.
- D. Guarantee zero defects.

Câu 245. Which analogy BEST represents the importance of having a software process:

- A. Taking an exam without studying.
- B. Cooking a meal by following a recipe.
- C. Listening to music while working.
- D. Shopping without a budget.

Câu 246. Which of the following is the most important reason why software process models differ across projects:

- A. Developer personal preferences.
- B. Differences in customer hobbies.
- C. Variation in project uncertainty and requirements stability.
- D. Availability of office equipment.

Câu 247. Early software processes were heavily influenced by:

- A. Social media platforms.
- B. Manufacturing and engineering disciplines.
- C. Modern agile startups.
- D. Open-source community collaboration.

Câu 248. A software process helps improve product quality primarily by:

- A. Standardizing development activities.
- B. Reducing team communication.
- C. Preventing all requirement changes.
- D. Eliminating the need for testing.

Câu 249. A process without defined stages or structure usually leads to:

- A. High predictability.
- B. Consistent product quality.
- C. Chaos and miscommunication.
- D. Reduced development time.

Câu 250. Which of the following is considered a major category of software process models:

- A. Sequential vs Functional.
- B. Predictive vs Adaptive.
- C. Horizontal vs Vertical.
- D. Internal vs External.

Câu 251. Which situation most strongly demands a structured software process:

- A. A small assignment done by one student.
- B. A large project with multiple teams and responsibilities.
- C. A fun side project made for entertainment.
- D. A quick prototype for personal learning.

Câu 252. One of the central challenges that drove the evolution of software processes was:

- A. Hardware prices decreasing.
- B. Increasing complexity and changing requirements.
- C. Fewer people wanting to be developers.
- D. Overabundance of technical documentation.

Câu 253. Which statement best describes the role of a process model:

- A. It guarantees the fastest possible development.
- B. It ensures that all teams worldwide use identical workflows.
- C. It provides a structured approach for producing software.
- D. It eliminates the need for planning.

Quiz02 · 24 câu

Câu 254. Predictive models rely heavily on:

- A. Early detailed planning.
- B. Flexible iteration.
- C. Minimal documentation.
- D. Constant requirement changes.

Câu 255. A predictive model is MOST effective when:

- A. Requirements are stable and well understood.
- B. Requirements change daily.
- C. Customers are unsure of what they want.
- D. The project requires rapid experimentation.

Câu 256. The Waterfall Model is BEST described as:

- A. A highly flexible and cyclic approach.
- B. A sequential process with fixed phases.
- C. A feedback-driven iterative model.
- D. A continuous flow system.

Câu 257. A key weakness of the Waterfall Model is that:

- A. It involves too much customer collaboration.
- B. It delivers working software late in the process.
- C. It prevents all requirement misunderstandings.
- D. It eliminates the need for documentation.

Câu 258. The V-Model primarily emphasizes:

- A. Minimum planning.
- B. Heavy prototyping.
- C. Early and structured test planning.
- D. No verification activities.

Câu 259. The "V" shape in the V-Model represents:

- A. Cost on the left, value on the right.
- B. Increasing complexity over time.
- C. Corresponding verification and validation phases.
- D. The velocity of development.

Câu 260. The Incremental Model differs from Waterfall because it:

- A. Eliminates documentation.
- B. Allows partial delivery of working software.
- C. Requires no upfront planning.
- D. Does not use iterations.

Câu 261. Predictive models tend to struggle when:

- A. Requirements are stable.
- B. System risk is low.
- C. Requirements change frequently.
- D. Projects demand extensive documentation.

Câu 262. What is a primary strength of predictive models:

- A. High flexibility during development.
- B. Early cost and schedule predictability.
- C. Encouraging frequent requirement changes.
- D. Avoiding all documentation.

Câu 263. The Waterfall Model is commonly used in:

- A. Projects with frequent change.
- B. Regulated industries needing strict documentation.
- C. Startups building experimental prototypes.
- D. Rapid innovation environments.

Câu 264. The IRS modernization failure highlights that Waterfall:

- A. Handles changing requirements well.
- B. Relying on early requirements can be risky.
- C. Eliminates the need for testing.
- D. Always completes on schedule.

Câu 265. Predictive models are MOST suitable when:

- A. Stakeholders cannot define requirements.
- B. The system must meet strict compliance standards.
- C. The customer prefers high adaptability.
- D. Requirements will evolve rapidly.

Câu 266. A major disadvantage of predictive models is that they:

- A. Provide too much customer collaboration.
- B. Lack flexibility for changing requirements.
- C. Require too little documentation.
- D. Rely heavily on working software early.

Câu 267. In the Waterfall Model, testing occurs:

- A. Only after the implementation phase is complete.
- B. Continuously throughout the project.
- C. Before requirements gathering.
- D. Before design is completed.

Câu 268. The Incremental Model is beneficial because it:

- A. Delivers value earlier than a single final release.
- B. Avoids all planning.
- C. Requires no customer interaction.
- D. Eliminates need for testing.

Câu 269. The V-Model is preferred in domains where:

- A. Rapid prototyping is essential.
- B. Safety and correctness are critical.
- C. Frequent requirement changes occur.
- D. Minimal documentation is acceptable.

Câu 270. The main assumption behind predictive models is that:

- A. Requirements will evolve organically.
- B. Requirements can be fully known early.
- C. Customers prefer flexible processes.
- D. Teams want minimal structure.

Câu 271. A negative outcome of rigid sequential phases is:

- A. Reduced clarity.
- B. Increased adaptability.
- C. Difficulty returning to earlier stages.
- D. Better handling of change.

Câu 272. One key reason predictive models rely on documentation is to:

- A. Remove the need for communication.
- B. Ensure traceability and compliance.
- C. Make coding unnecessary.
- D. Allow spontaneous changes.

Câu 273. When using Waterfall, customers typically see the working product:

- A. Very early.
- B. Only near the end.
- C. After each daily iteration.
- D. At the start of each sprint.

Câu 274. Predictive planning involves:

- A. Planning only small increments at a time.
- B. Deferring planning until after coding begins.
- C. Establishing detailed schedules early.
- D. Adjusting plans continuously.

Câu 275. Which project is LEAST suited for predictive models:

- A. A startup app exploring uncertain features.
- B. Hospital patient records system.
- C. Banking compliance platform.
- D. Government defense control software.

Câu 276. The biggest challenge predictive models face is:

- A. Handling volatile requirements.
- B. Creating documentation.
- C. Managing test cycles.
- D. Organizing teams.

Câu 277. The core philosophy of predictive models is:

- A. Responding to change over following a plan.
- B. Building working software before defining requirements.
- C. Planning thoroughly before development begins.
- D. Prioritizing collaboration over documentation.

Quiz03 · 30 câu

Câu 278. Adaptive models are designed to:

- A. Prevent requirement changes.
- B. Accommodate evolving requirements.
- C. Replace communication with documentation.
- D. Use no iterations.

Câu 279. A key characteristic of adaptive processes is:

- A. Strict phase gating.
- B. Detailed upfront planning.
- C. Iterative and feedback-driven development.
- D. Heavy documentation focus.

Câu 280. Which statement best describes the Product Backlog in Scrum:

- A. A living, prioritized list of everything needed in the product.
- B. A fixed contract signed at project start.
- C. A list of tasks for developers to complete daily.
- D. A complete design document.

Câu 281. The Sprint Backlog is BEST described as:

- A. The set of Product Backlog items selected for the current sprint.
- B. A list of all work in the entire project.
- C. A document controlled only by the Product Owner.
- D. A high-level requirement specification.

Câu 282. The Spiral Model primarily focuses on:

- A. Eliminating all project risks.
- B. Risk-driven development cycles.
- C. Minimizing customer involvement.
- D. Rapid, unstructured delivery.

Câu 283. RUP (Rational Unified Process) is best described as:

- A. A pure predictive model.
- B. A purely adaptive model with no structure.
- C. A structured yet iterative framework.
- D. A model that eliminates documentation.

Câu 284. Agile was created because developers were frustrated with:

- A. Too much customer feedback.
- B. Heavy, rigid, inefficient processes.
- C. Insufficient documentation.
- D. Lack of automated tools.

Câu 285. The Agile Manifesto emphasizes:

- A. Following a fixed plan.
- B. Maximizing documentation.
- C. Individuals and interactions.
- D. Minimizing customer collaboration.

Câu 286. Which Agile value focuses on prioritizing customer needs:

- A. Responding to change over following a plan.
- B. Working software over comprehensive documentation.
- C. Customer collaboration over contract negotiation.
- D. Individuals and interactions over processes and tools.

Câu 287. A key principle of Agile is:

- A. Delivering software only once at the end.
- B. Avoiding iterative development.
- C. Welcoming changing requirements.
- D. Releasing no working software until the entire project is finished.

Câu 288. Scrum is best described as:

- A. A rigid sequential methodology.
- B. A lightweight Agile framework.
- C. A predictive project management tool.
- D. A design-only technique.

Câu 289. The Product Owner is primarily responsible for:

- A. Removing team obstacles.
- B. Writing all code.
- C. Managing and prioritizing the Product Backlog.
- D. Conducting the Daily Scrum.

Câu 290. The Scrum Master's core responsibility is to:

- A. Prioritize backlog items.
- B. Facilitate Scrum practices and remove blockers.
- C. Approve design decisions.
- D. Write user stories.

Câu 291. The Development Team in Scrum is:

- A. Self-managing and cross-functional.
- B. Strictly controlled by the Scrum Master.
- C. Not allowed to collaborate.
- D. Forbidden from talking to the Product Owner.

Câu 292. Each Sprint produces:

- A. A fully completed project.
- B. A user manual.
- C. A potentially shippable product increment.
- D. A complete design document.

Câu 293. The Daily Scrum is intended to:

- A. Report to the Scrum Master.
- B. Identify progress and blockers.
- C. Review the entire project.
- D. Conduct testing activities.

Câu 294. The Sprint Review focuses on:

- A. Reflecting on team performance.
- B. Demonstrating the increment and gathering feedback.
- C. Planning the next sprint.
- D. Estimating user stories.

Câu 295. The Sprint Retrospective focuses on:

- A. Showing the increment.
- B. Reviewing the Product Backlog.
- C. Identifying process improvements.
- D. Finalizing the product release.

Câu 296. Kanban is best described as:

- A. A time-boxed Agile framework.
- B. A continuous flow method visualizing work.
- C. A predictive model.
- D. A debugging technique.

Câu 297. Extreme Programming (XP) emphasizes:

- A. Heavy documentation.
- B. Individual contributors working alone.
- C. Technical excellence such as TDD and pair programming.
- D. Waterfall-style delivery.

Câu 298. Lean Software Development focuses on:

- A. Adding extra steps to ensure safety.
- B. Eliminating waste and delivering fast.
- C. Avoiding customer involvement.
- D. Maximizing documentation.

Câu 299. FDD (Feature-Driven Development) is especially suited for:

- A. Small, uncertain prototypes.
- B. Very large, domain-rich systems.
- C. Systems with no clear features.
- D. Teams that avoid documentation.

Câu 300. Scaled Agile Frameworks (e.g., SAFe) exist because:

- A. Agile cannot work in any organization.
- B. Large enterprises need coordination across many teams.
- C. Agile requires minimal planning.
- D. Predictive models scale automatically.

Câu 301. Which scenario BEST suits Agile:

- A. Government defense system with strict compliance.
- B. Building a medical device requiring formal approvals.
- C. A startup building a new social media app.
- D. Financial reporting software with fixed requirements.

Câu 302. Agile welcomes changing requirements because:

- A. It avoids planning.
- B. Customers never know what they want.
- C. Feedback-driven iterations improve product alignment.
- D. Documentation is not important.

Câu 303. The core unit of work in Scrum is:

- A. The requirement document.
- B. The Sprint.
- C. The Gantt chart.
- D. The phase gate.

Câu 304. A major advantage of Agile over predictive models is:

- A. It eliminates testing.
- B. It adapts to change more easily.
- C. It requires no planning.
- D. It guarantees early cost accuracy.

Câu 305. Scrum teams should be:

- A. Large with specialized roles.
- B. Self-organizing with cross-functional skills.
- C. Managed hierarchically with command-style control.
- D. Focused mainly on documentation.

Câu 306. A Scrum Sprint is typically:

- A. A flexible time period that changes weekly.
- B. A fixed-length iteration with clear goals.
- C. A continuous flow of work.
- D. A meeting with stakeholders.

Câu 307. The Agile mindset is fundamentally about:

- A. Following strict rules.
- B. Avoiding planning.
- C. Embracing learning and change.
- D. Reducing team collaboration.

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Câu 308. When creating a persona to represent a product user, what is the primary purpose of including personal details like a name, age, and photograph?

- A. To ensure the development team can perform a background check on the user type.
- B. To create a realistic character that helps the team empathize with the user ("step into their shoes").
- C. To meet the legal requirements for user documentation.
- D. To define the specific salary bracket for pricing the software.

Câu 309. Which of the following is the most effective element to include in a persona description to help developers judge UI complexity?

- A. The user's long-term life goals (e.g., "wants to retire early").
- B. The user's educational background and technical skill level.
- C. The user's favorite color scheme.
- D. The user's manager's name.

Câu 310. Why is it generally advised to avoid focusing on "user goals" (e.g., "career progression") when writing personas?

- A. Users never have goals when using software.
- B. Goals are too broad and often do not directly relate to why a user needs a specific piece of software to get through their day.
- C. Goals are always financial and irrelevant to functionality.
- D. It is impossible to write a goal in less than three paragraphs.

Câu 311. You are developing a new product and have very little data. You create "proto-personas" based on the team's best guesses. What is the limitation of this approach?

- A. They are too detailed to be useful.
- B. They cannot be used to write scenarios.
- C. They are not as accurate as those based on user research, though they still provide a shared team vision.
- D. They require legal approval before use.

Câu 312. How many personas should you typically create to identify the key features of a system effectively?

- A. One for every potential user of the system.
- B. At least 20 to cover every possible demographic.
- C. A small number (e.g., up to 5) to avoid overlapping requirements and confusion.
- D. Only one "average" persona.

Câu 313. Which of the following best describes the correct "level of detail" for a starting scenario?

- A. A strict, step-by-step list of every mouse click and keystroke.
- B. A high-level narrative describing a user's problem and an imagined solution without technical specification.
- C. A complete code-level specification of the feature.
- D. A vague statement saying "the user solves the problem" with no description of how.

Câu 314. When writing a scenario, why might you include specific technical details (like "Log in with Google") rather than a generic phrase (like "User performs authentication")?

- A. Because generic phrases can be confusing, while specific examples are often easier for users and developers to understand.
- B. Because you are required to use Google products.
- C. Because the code has already been written.
- D. Because scenarios must always dictate the final implementation database.

Câu 315. What is the primary advantage of using a "narrative" style (natural language) for scenarios over a "structured" style (fields and flowcharts)?

- A. Narrative scenarios are machine-readable.
- B. Narrative scenarios are accessible and understandable to non-technical users and stakeholders.
- C. Structured scenarios are too short.
- D. Narrative scenarios are legally binding contracts.

Câu 316. Which action should you take to identify software features from a written scenario?

- A. Count the number of adjectives used.
- B. Highlight the "active verbs" (actions) in the text to see what the user is doing (e.g., "upload," "share").
- C. Rewrite the scenario in a different language.
- D. Remove all references to the persona.

Câu 317. A scenario describes a teacher using a tool to organize a class trip. What represents the "Problem" element of this scenario?

- A. The teacher logs in.
- B. The teacher wants to share photos but needs to ensure student privacy and copyright compliance.
- C. The teacher has a degree in History.
- D. The system sends an email.

Câu 318. Scenarios are described as "tools for thinking." What does this imply about their completeness?

- A. They must cover every single edge case and error state.
- B. They must be perfect before any coding begins.
- C. They do not have to be complete or consistent; their job is to stimulate design creativity.
- D. They replace the need for testing.

Câu 319. Which of the following follows the standard format of a User Story?

- A. "The system shall allow the user to print documents."
- B. "As a , I want to , so that ."
- C. "Clicking the blue button saves the file to the cloud."
- D. "Jack, aged 32, needs a way to save time."

Câu 320. What is the main purpose of the "So that..." clause (the rationale) in a User Story?

- A. To increase the word count of the documentation.
- B. To help developers understand why a feature is needed, potentially triggering alternative ideas for implementation.
- C. To tell the user how to perform the task.
- D. To assign the task to a specific developer.

Câu 321. You have a User Story that says: "As a system manager, I need to back up the entire system and restore files." This story is very complex and will take weeks to build. What is this called?

- A. A Bug.
- B. An Epic.
- C. A Persona.
- D. A Scenario.

Câu 322. Why should you generally avoid writing "negative" user stories (e.g., "As a user, I don't want...") for a product backlog?

- A. It is impossible to write system tests that demonstrate a negative (that something didn't happen).
- B. Agile methodologies forbid negative language.
- C. Developers prefer positive reinforcement.
- D. Negative stories are too short.

Câu 323. How do User Stories differ from Scenarios regarding "context"?

- A. Stories provide more context than scenarios.
- B. Scenarios provide rich context about normal work and goals, while stories are finer-grain statements of a single need.
- C. Stories and scenarios are identical in context.
- D. Stories describe the user's background; scenarios describe the system's code.

Câu 324. What is the correct chronological flow when using these techniques to design a product?

- A. Features -> Personas -> Scenarios -> Stories
- B. Personas -> Scenarios -> Stories -> Features
- C. Stories -> Personas -> Features -> Scenarios
- D. Scenarios -> Features -> Personas -> Stories

Câu 325. You are analyzing a scenario where a teacher "creates a group web page." Which specific artifact would you write to help plan the implementation of this specific action?

- A. A new Persona for the web page.
- B. A User Story (e.g., "As a teacher, I want to create a web page...").
- C. A marketing plan.
- D. A new Scenario about the web page.

Câu 326. When identifying features from scenarios, you notice a user needs to "log in" in three different scenarios. What feature design principle suggests you should create one single "Authentication" feature rather than three separate ones?

- A. Coherence (The feature should be linked to a single item of functionality).
- B. Novelty (The feature should be new).
- C. Complexity (The feature should be difficult).
- D. Independence (The feature should not work with others).

Câu 327. If you are writing a User Story based on a scenario where a user needs to ensure their data isn't shared, how should you rephrase the negative constraint "I don't want my data shared"?

- A. "As a user, I want to control the information transmitted to external servers so I can ensure privacy."
- B. "As a user, I want to ban all data sharing."
- C. "As a user, I don't want the system to work."
- D. "As a user, I want to delete the application."

ĐÁP ÁN

Đáp án được sắp xếp theo thứ tự câu hỏi trong từng tuần. Câu có nhiều đáp án đúng được ghi đầy đủ (ví dụ: A, B).

TUẦN 01 — Git & Version Control

#1	#2	#3	#4	#5	#6
C	B	B	B	B	C
#7	#8	#9	#10	#11	#12
C	D	B	B	B	C
#13	#14	#15	#16	#17	#18
A	B	B	B	B	C
#19	#20	#21	#22	#23	#24
B	C	B	B	B	C

TUẦN 02 — Software Testing

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A	D	B	C	C	C
#31	#32	#33	#34	#35	#36
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#37	#38	#39	#40	#41	#42
C	B	C	D	A, B	B
#43	#44	#45	#46	#47	#48
D	C	C	D	B	A
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#55	#56	#57	#58	#59	#60
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C	B	C	C	C	C
#79	#80	#81	#82	#83	#84
C	B	B	A	D	C
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A	B	B	B	B	

TUẦN 03 — Input Validation & Debugging

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A	C	A	C	A	A
#108	#109	#110	#111		
A	A	C	B		

TUẦN 04 — Software Architecture

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#136	#137	#138			
B	B	B			

TUẦN 05 — Cloud & Microservices

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TUẦN 06 — Requirements Engineering

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TUẦN 07 — Software Process Models

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#274	#275	#276	#277	#278	#279
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C	C	B	C	B	A
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C	B	B	C	B	C
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#304	#305	#306	#307		
B	B	B	C		

TUẦN 08 — User Personas & UX

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#314	#315	#316	#317	#318	#319
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#320	#321	#322	#323	#324	#325
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#326	#327				
A	A				